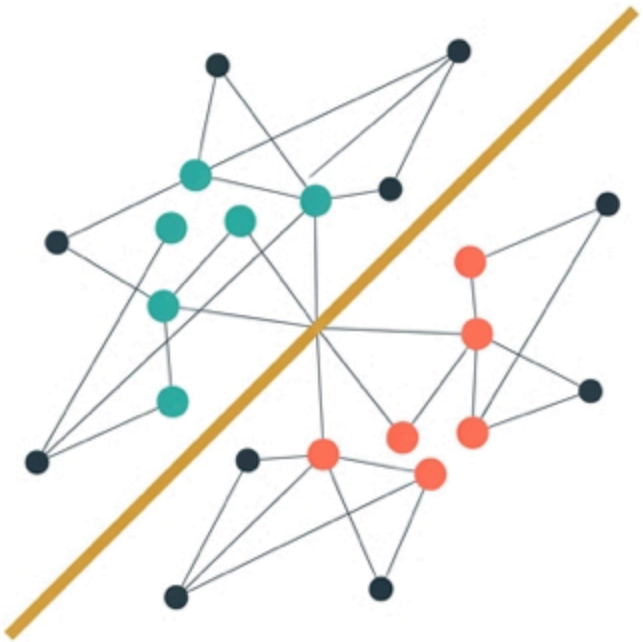
Python 機器學習入門

第九章：Scikit-Learn 與機器學習基礎專題

講師：Python 程式設計教學團隊

The Machine Learning Toolkit

A visual taxonomy of algorithms, evaluation metrics, and predictive pipelines in Scikit-Learn.

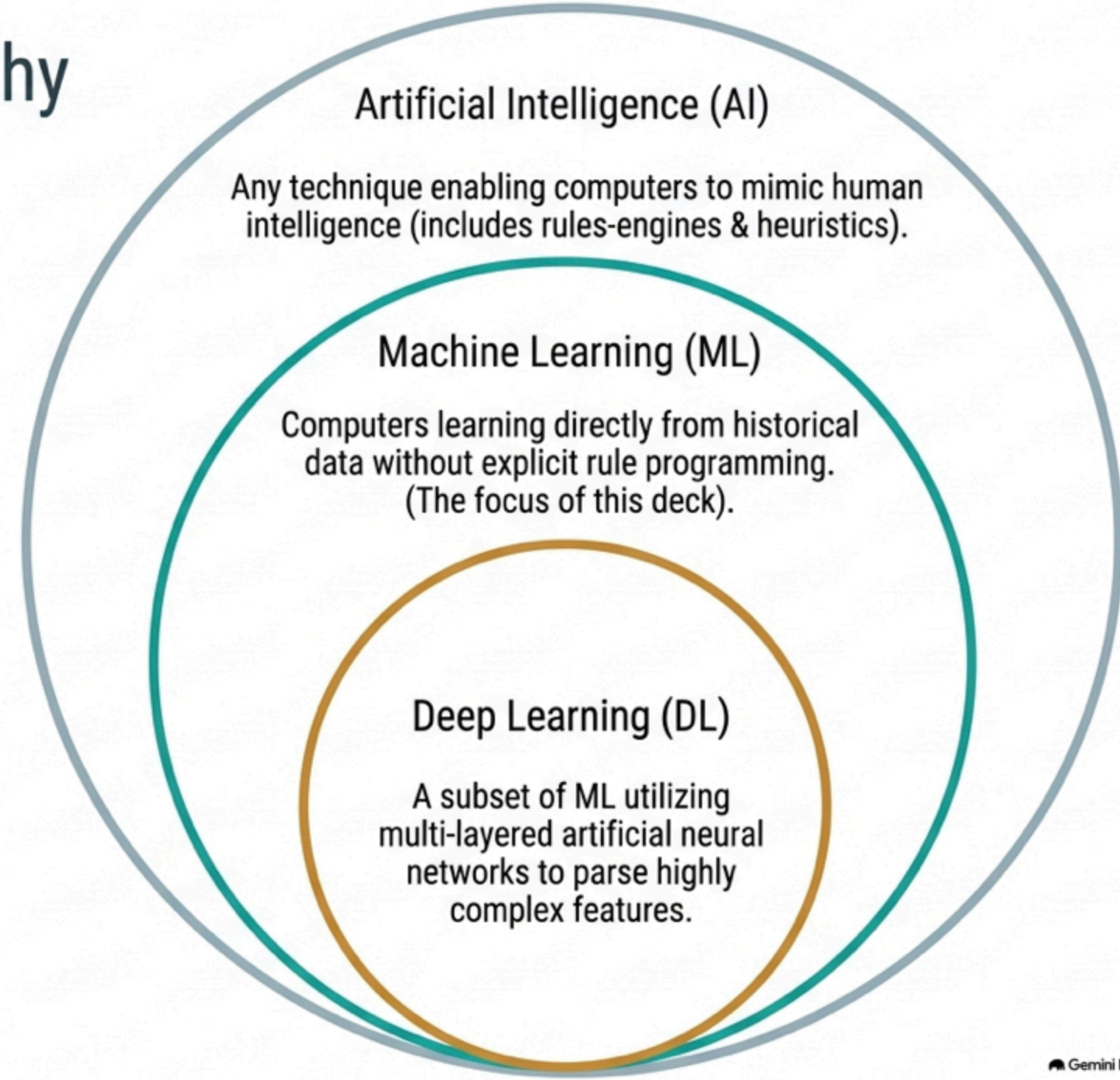


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9.1 機器學習基礎觀念

- **AI vs ML vs DL**：
  - **AI** (人工智慧)：模擬人類智慧。
  - **ML** (機器學習)：從歷史數據中「自我學習」規律。
  - **DL** (深度學習)：利用多層神經網路處理複雜非線性結構。
- **監督式學習 (Supervised)**：
  - 有給定標籤。如：**分類**（預測類別）與**迴歸**（預測數值）。
- **非監督式學習 (Unsupervised)**：
  - 無給定標籤。如：**分群**。

The Intelligence Hierarchy



Core Concept: Machine Learning marks the transition from explicit programming to pattern inference.

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The Three Learning Paradigms



Supervised Learning

- Data: Features (X) + Labels (y)
- Goal: Predict outcomes based on historical examples.
- Sub-types: Classification (discrete labels) & Regression (continuous values).



Unsupervised Learning

- Data: Features (X) only, no labels.
- Goal: Discover hidden structures, patterns, or groupings.
- Sub-types: Clustering (grouping similar samples), Dimensionality Reduction.



Reinforcement Learning

- Data: Interactive Environment.
- Goal: Train agents via trial, error, and reward mechanisms.
- Examples: AlphaGo, autonomous robotics, game theory.

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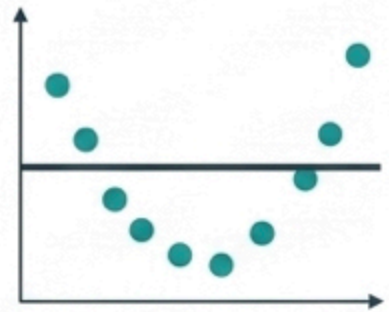
9.2 監督式學習：分類任務與超參數調校

- **K-近鄰演算法 (KNN)**：
  - 依據最近的 K 個鄰居的多數決決定類別。
  - 核心數學：歐氏距離與曼哈頓距離。
- **決策樹 (Decision Tree)**：
  - 利用「二分問答」將資料分類，追求最純淨分割。
  - 核心指標：吉尼不純度 (Gini Impurity)。
- **隨機森林 (Random Forest)**：
  - 集成學習。利用 Bagging 與特徵隨機選擇，統合多個決策樹降低過擬合。

The Fundamental Law: The Bias-Variance Tradeoff

High Bias / Low Variance

Underfitting



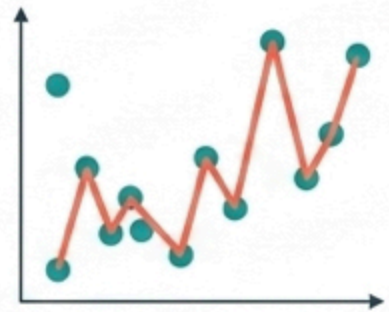
Model is too simple. Fails to capture the underlying structure.

The Balance Point

Minimum Total Error.
The optimal model complexity.

Low Bias / High Variance

Overfitting



Model is too complex. Memorizes random noise, failing on new test data.

Total Error = Bias + Variance + Irreducible Noise

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Supervised Toolkit: K-Nearest Neighbors (KNN)

The Mechanism

A non-parametric algorithm that predicts new samples via a majority vote of its K closest neighbors in feature space.

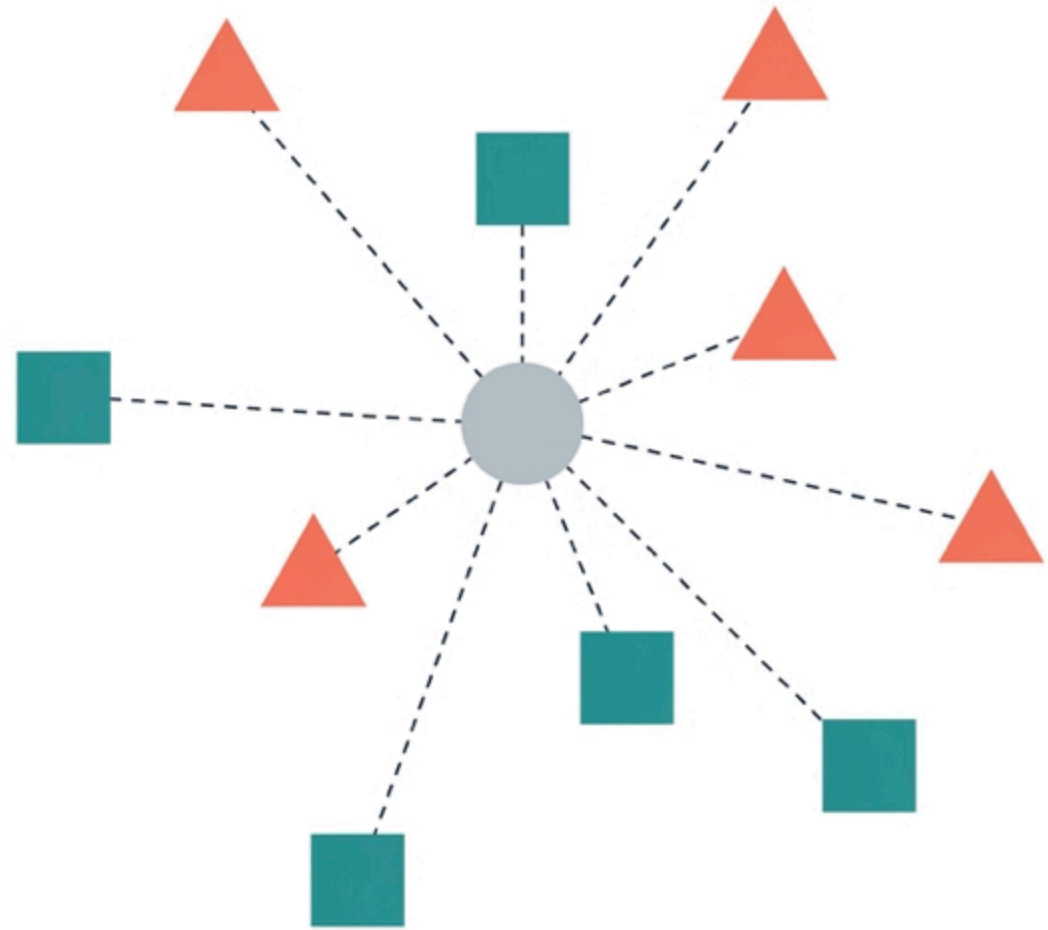
Distance Metrics

Euclidean Distance:

$$d(\mathbf{p}, q) = \sqrt{\sum (p_i - q_i)^2}$$

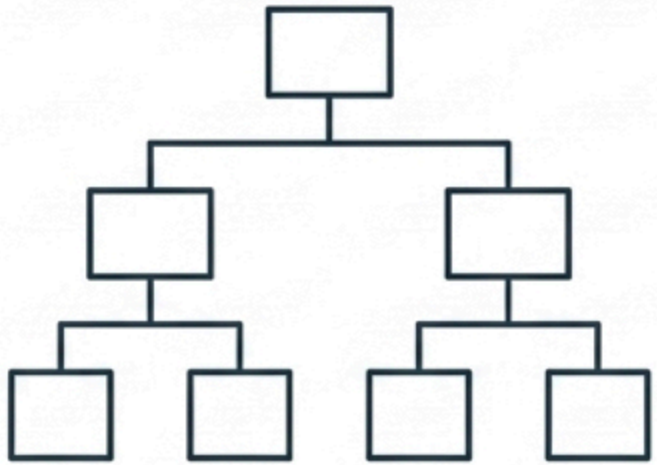
Manhattan Distance:

$$d(\mathbf{p}, q) = \sum |p_i - q_i|$$



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Supervised Toolkit: Decision Trees & The Random Forest

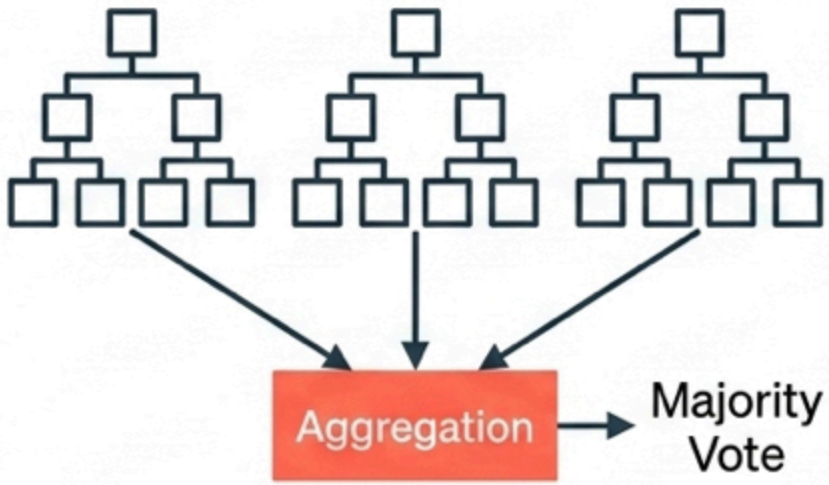


Decision Tree Logic

Splits data sequentially by minimizing Gini Impurity:

$Gini = 1 - sum(P_i^2)$

Danger: Setting max_depth to None causes infinite splitting, pure leaves, and extreme overfitting (High Variance).

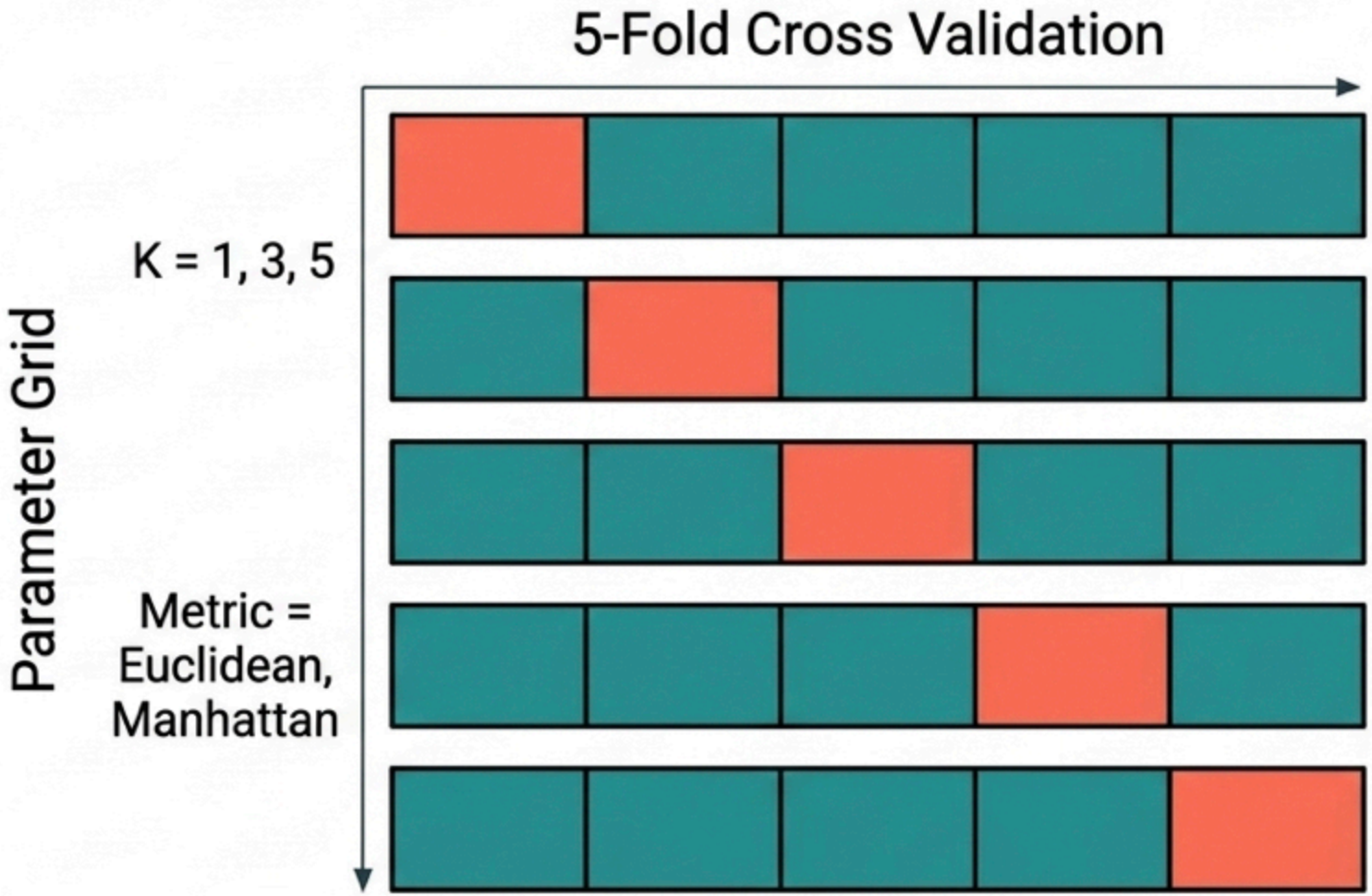


The Random Forest (Ensemble)

- **Ensemble Learning:** Combining multiple weak learners into a single strong predictor.
- **Bagging (Bootstrap Aggregating):** Each tree is trained on random subsets of data with replacement.
- **Feature Randomness:** Node splits evaluate only random subsets of features to prevent single dominant features from controlling every tree.

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“Optimizing the Engine: Grid Search & Cross-Validation” in Deep Slate



1. Exhaustive Search

GridSearchCCV mechanically tests every possible combination in the defined parameter grid.

2. Cross-Validation

Each combination is evaluated across different folds of the data to prevent single-slice bias.

3. Optimal Output

Automatically selects and returns the single parameter configuration with the highest generalized accuracy.

Gemini Notebook

網格搜尋參數調優與交叉驗證代碼

```
from sklearn.model_selection import GridSearchCV
from sklearn.neighbors import KNeighborsClassifier

# 設定搜尋超參數網格
knn_param_grid = {
    'n_neighbors': [1, 3, 5, 7, 9],
    'weights': ['uniform', 'distance'],
    'metric': ['euclidean', 'manhattan']
}

# 5-fold 交叉驗證網格搜尋
grid_search = GridSearchCV(KNeighborsClassifier(), knn_param_grid, cv=5, scoring='accu
grid_search.fit(X_train, y_train)

print(f"最佳參數組合：{grid_search.best_params_}")
print(f"交叉驗證最佳精度：{grid_search.best_score_:.4f}")
```

Concept Check Question (CCQ 1)

在機器學習中，使用 `GridSearchCV` 進行「超參數網格搜尋與交叉驗證」的主要目的為何？

- **A.** 為了加速模型訓練的速度，避免使用 CPU
- **B.** 自動在各種參數組合中，找出最能防止過擬合且泛化能力最佳的參數設定
- **C.** 為了將無標籤的資料集進行自動分群
- **D.** 將特徵維度進行降維以利於繪圖



CCQ 1 - 答案與解析

正確答案：B. 自動在各種參數組合中，找出最能防止過擬合且泛化能力最佳的參數設定



- **解析**：
  - `GridSearchCV` 採用窮舉法測試網格內的參數，配合 K-折交叉驗證以防範單次切分所產生的預測偏誤，進而決定出泛化效能最好的超參數，故選 B。

Concept Check Question (CCQ 2)

當決策樹（Decision Tree）的 `max_depth` (最大深度) 參數設定為 `None`（不限制深度）時，模型最易面臨什麼風險？

- **A.** 模型會因為結構過於簡單而產生欠擬合 (Underfitting)
- **B.** 決策樹會無法進行多類別分類
- **C.** 決策樹會過度分裂，極易產生過擬合 (Overfitting) 並喪失泛化能力
- **D.** 程式會因為死迴圈而當機



CCQ 2 - 答案與解析

正確答案：C. 決策樹會過度分裂，極易產生過擬合 (Overfitting) 並喪失泛化能力



- **解析**：
  - 不設最大深度的樹會一直分裂到所有葉節點都只包含單一類別（即 100% 純淨），這會使其記住訓練集中微小的隨機噪聲，降低新樣本預測能力。
  - 通常需要透過限制樹深度（Pre-pruning）來做正規化，故選 C。

9.3 監督式學習：迴歸任務與評估指標

- **多元線性迴歸 (OLS)**：
  - 藉由多項特徵求解預測值：$\hat{y} = w_1x_1 + w_2x_2 + w_3x_3 + b$。
  - 核心算法為最小平方法，旨在最小化殘差平方和。
- **L1 & L2 正規化 (Regularization)**：
  - **脊迴歸 (Ridge / L2)**：加上權重平方懲罰項，均勻縮小所有權重。
  - **套索迴歸 (Lasso / L1)**：加上權重絕對值懲罰，強制將不重要權重歸 0。
- **評估指標**：
  - MAE (直觀平均誤差)、MSE (加重懲罰大誤差) 與 R^2 Score (解釋度)。

Supervised Toolkit: Multiple Linear Regression

$$y = w_1x_1 + w_2x_2 + w_3x_3 + b$$

Predicted Target
(e.g., House Price)

Learned Weights
(Coefficients)

Features
(Size, Rooms, Age)

Intercept
(Base offset)

Core Mechanism:

Uses Ordinary Least Squares (OLS) to find the specific weights that minimize the squared differences between predictions and actual observed values.

Example Output:

Predicted Price = 40(Size) + 150(Rooms) - 5(Age) + 100

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Controlling Complexity: Regularization (L1 vs L2)

Ridge Regression (L2 Penalty)

$$\text{Loss} = \text{OLS} + \alpha * \sum(w_j^2)$$

- **Mechanism:** Penalizes the square of the weights.
- **Effect:** Shrinks all weights evenly, but rarely forces them to absolute zero.
- **Use Case:** Ideal for handling highly correlated features without discarding any.

Lasso Regression (L1 Penalty)

$$\text{Loss} = \text{OLS} + \alpha * \sum(|w_j|)$$

- **Mechanism:** Penalizes the absolute value of the weights.
- **Effect:** Forces less important feature weights to exactly 0.
- **Use Case:** Acts as built-in Feature Selection, simplifying the model.

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Measuring Success: Regression Metrics

Mean Absolute Error (MAE)

$1/n * \sum(| y - y_pred |)$

The average physical distance between prediction and reality. Highly intuitive and robust to outliers.

Mean Squared Error (MSE)

$1/n * \sum((y - y_pred)^2)$

Squares the errors before averaging. Extremely sensitive to massive outliers (penalizes large errors heavily).

R-Squared Score (R^2)

$1 - (SS_res / SS_tot)$

The Coefficient of Determination. An R^2 of 0.85 means 85% of the variance in the target variable is explained by the model.

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Concept Check Question (CCQ 3)

在評估迴歸預測模型效能時，若我們算出模型的決定係數 R^2 值為 `0.85`，這代表什麼工程含義？

- **A.** 該模型預測對了 85% 的數值，剩下的 15% 數值預測錯誤
- **B.** 該模型所預測的值比真實值平均貴了 85 單位
- **C.** 自變數（特徵）能夠解釋因變數（標籤）中 85% 的變異量
- **D.** 模型有 85% 的機率會產生過擬合



CCQ 3 - 答案與解析

正確答案：C. 自變數（特徵）能夠解釋因變數（標籤）中 85% 的變異量



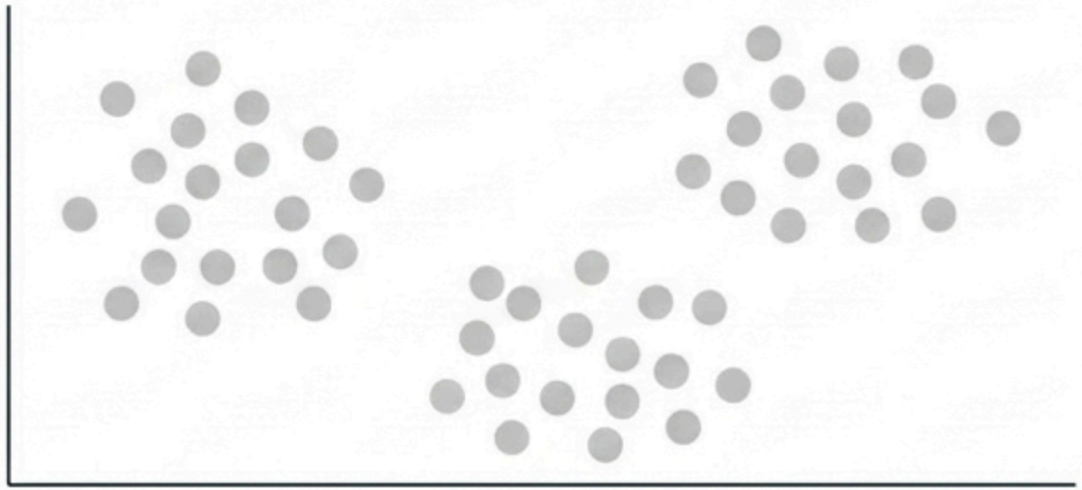
- **解析**：
  - R^2 (決定係數) 反映模型相對於平均基準預測的解釋力。
  - 0.85 代表 85% 的變異已包含於迴歸曲線中。R^2 越高代表模型擬合效果越好，故選 C。

9.4 非監督式學習：K-Means 資料分群與肘部法

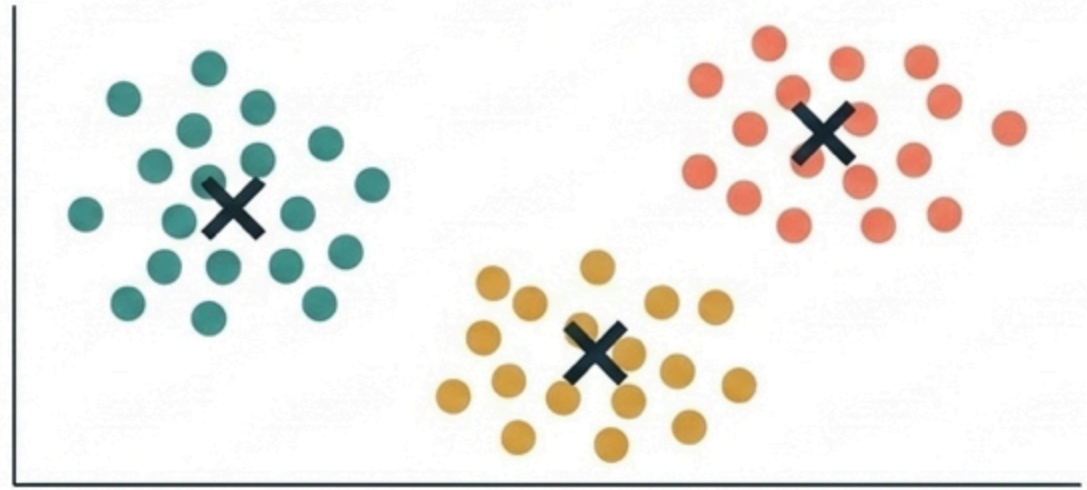
- **K-Means 聚類演算法**：
  - 重複計算群心並更新資料點分配，目標是最小化點至群心之距離和。
- **肘部法 (Elbow Method)**：
  - 計算不同 K 值對應之 Inertia (群內誤差平方和)。
  - 在折線圖上手肘彎折的轉折點即為最合理的 K 值選擇。
- **K-Means 限制**：
  - 必須預先指定分群數。
  - 假設資料呈現球狀圓形分佈。新月形或非均質密度資料分群建議採用 **DBSCAN**。

Unsupervised Toolkit: K-Means Clustering

Input: Unlabeled Data



Output: Clustered Data



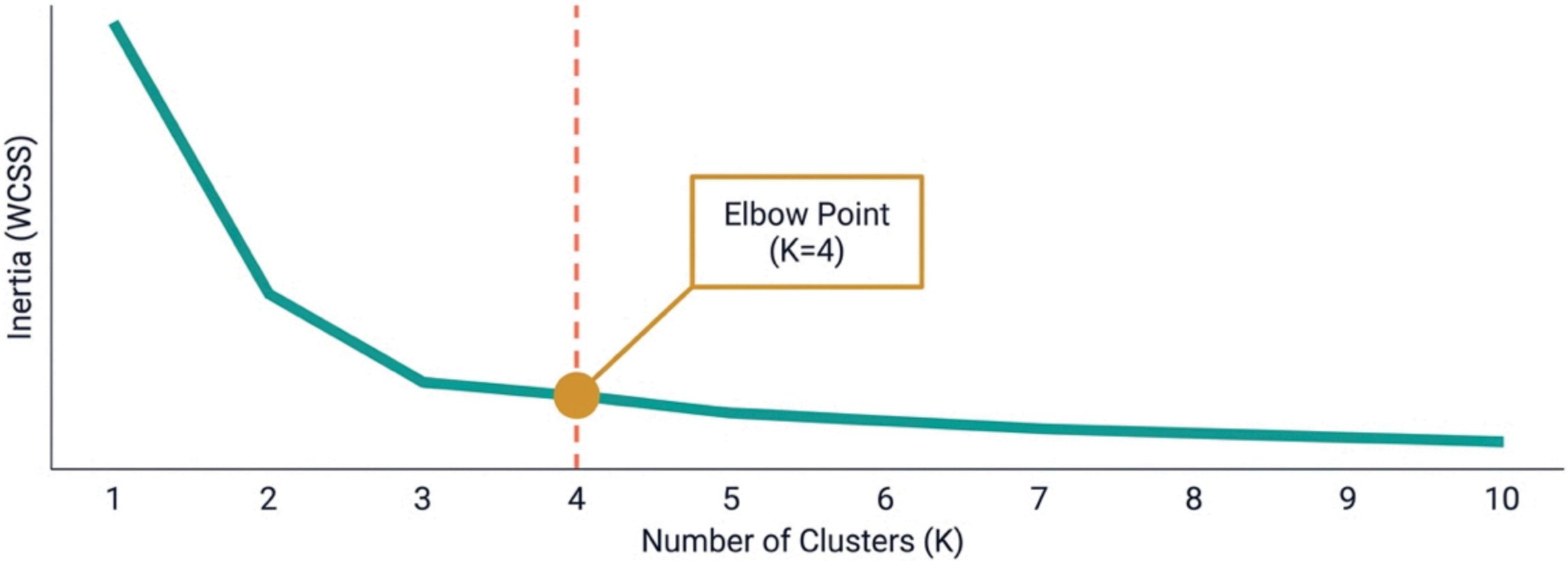
How it Works:
Assigns data points to the nearest centroid, then recalculates the centroid position based on the mean of the assigned points. Iterates until stable.

Critical Limitations:
1. Assumes data is grouped in spherical, similarly-sized clusters.
2. The user must manually define K in advance.
3. Highly sensitive to initial centroid placement.

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Finding the Balance: The Elbow Method

$$\text{Inertia (WCSS)} = \sum_i \left(\sum_i \sum_j \left(|x_i - u_j|^2 \right) \right)$$



The elbow represents the point of diminishing returns. After this threshold, increasing the number of clusters reduces intra-cluster variance by only marginal, insignificant amounts.

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Concept Check Question (CCQ 4)

在實施 K-Means 分群時，使用「肘部法」尋找最佳分群數，下列何者正確？

- **A.** 轉折點代表 Inertia 開始變為負值的地方
- **B.** 轉折點代表在此群數之後，增加群數所能降低的群內誤差和幅度明顯變小
- **C.** 轉折點代表分群準確度達到 100%
- **D.** 轉折點後的 K 值代表模型開始欠擬合



CCQ 4 - 答案與解析

正確答案：B. 轉折點代表在此群數之後，增加群數所能降低的群內誤差和幅度明顯變小



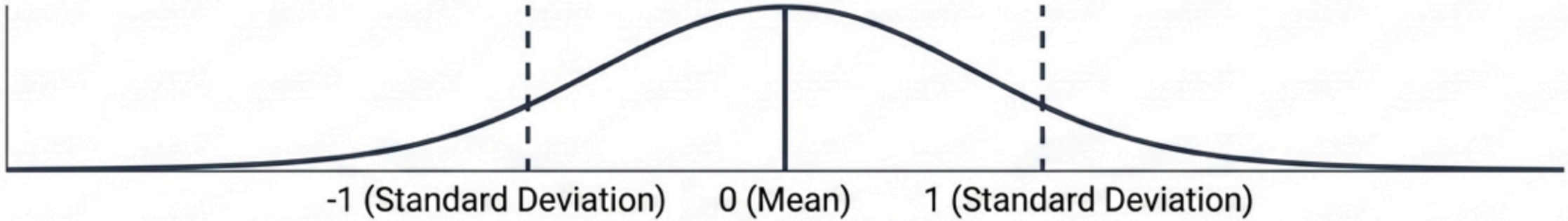
- **解析**：
  - 在肘部彎曲之前，增加 K 值能大幅縮小 Inertia ；彎折點之後，誤差縮減速度明顯放緩，此處為邊際效應最佳折衷點，故選 B。

9.5 關鍵觀念與特徵工程 (Feature Engineering)

- **特徵縮放 (Feature Scaling)**：
  - **StandardScaler** (Z-Score)：縮放為平均 0、標準差 1。適用於距離型算法 (KNN, SVM) 與梯度下降法。
  - **MinMaxScaler**：壓縮至 0 ~ 1。
- **獨熱編碼 (One-Hot Encoding)**：
  - 用於將無順序關係的類別特徵轉換為二進位獨立向量。
  - 防止模型假設類別特徵具有大小數值大小偏見。

The Engine Room: Feature Standardization

StandardScaler (Z-Score)



Centers data around a mean of 0 with a variance of 1. Robust against extreme values.

MinmxScaler



Proportionally compresses all data strictly between 0 and 1. Highly sensitive to outliers.

Why it Matters: Distance-based algorithms (like KNN) will artificially prioritize features with larger numerical magnitudes if they are not scaled to a unified dimensional space.

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The Engine Room: One-Hot Encoding

Before

City
Taipei
Taichung
Kaohsiung

pd.get_dummies()

After - One-Hot Encoded

City_Taipei	City_Taichung	City_Kaohsiung
1	0	0
0	1	0
0	0	1



Danger: Encoding as 1, 2, 3 tricks the model into assuming ordinal hierarchy or distance.

Core Principle: One-Hot Encoding creates orthogonal, independent dimensions for categorical variables, entirely eliminating numerical bias.

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Concept Check Question (CCQ 5)

在處理無序「類別特徵（如科系、班級）」時，為什麼通常建議使用 One-Hot Encoding 而非直接數字對照（如電機=1, 機械=2）？

- **A.** 因為 Scikit-Learn 的模型只支援 0 或 1 的輸入
- **B.** 避免模型錯誤地假設這些類別特徵之間存在數值順序或倍數關係
- **C.** One-Hot Encoding 可以自動刪除重複的特徵
- **D.** 整數對照編碼會佔用十倍以上的記憶體



CCQ 5 - 答案與解析

正確答案：B. 避免模型錯誤地假設這些類別特徵之間存在數值順序或倍數關係

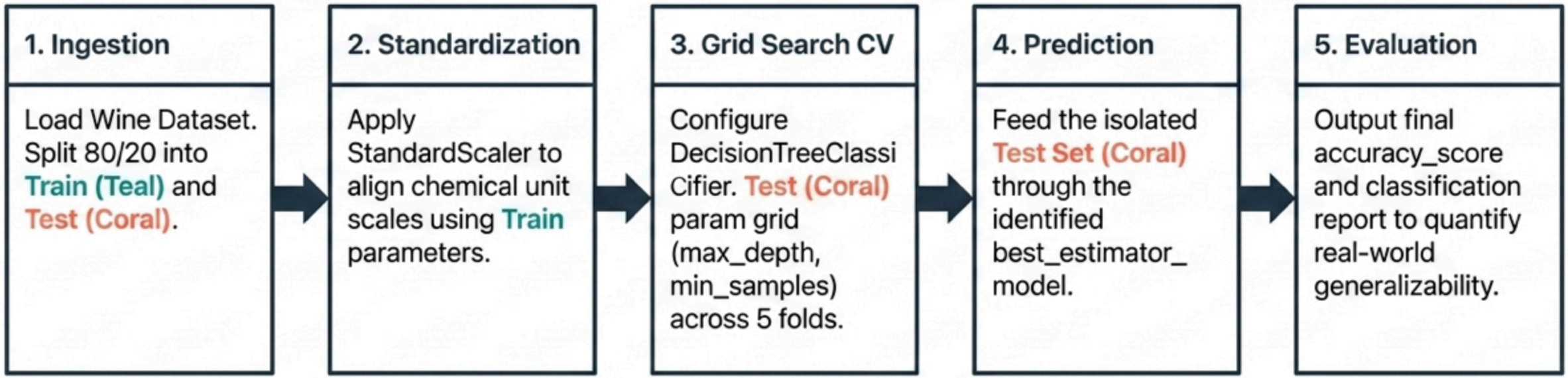


- **解析**：
  - 若寫入 $1, 2, 3$，模型會計算距離並假設科系可以進行數學加減運算。
  - One-Hot 編碼透過將各選項對應到獨立維度，使它們彼此正交，維持合理的類別間關係，故選 B。

9.6 本章綜合實作專題

- **紅酒多重分類器任務**：
  - 使用 `StandardScaler` 標準化。
  - 建立 `DecisionTreeClassifier` 分類器。
  - 使用 `GridSearchCV` 進行 5 折交叉驗證。
  - 使用 `classification_report` 評估 Precision, Recall 與 F1-score。

Synthesis: The Capstone Pipeline



Gemini Notebook